# Target Pipeline EDA and Analysis with PySpark

Run the pipeline first so the files under `output/` exist, then run this notebook top to bottom. All EDA and analysis computations below use a PySpark DataFrame; pandas is only used to plot the already-aggregated Spark results.

In [4]:
```
from pathlib import Path
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
from pyspark.sql import SparkSession
from pyspark.sql import functions as F

sns.set_theme(style="whitegrid")
plt.rcParams["figure.figsize"] = (12, 6)
pd.set_option("display.max_rows", 200)
pd.set_option("display.float_format", lambda value: f"{value:,.2f}")

PROJECT_ROOT = Path.cwd().resolve().parent if Path.cwd().name == "notebooks"
OUTPUT_DIR = PROJECT_ROOT / "output"
CLEANED_DATA_PATH = OUTPUT_DIR / "cleaned__target_data.csv"
CITY_REPORT_DIR = OUTPUT_DIR / "sales_by_city_report"

if not CLEANED_DATA_PATH.exists():
    raise FileNotFoundError("Run the pipeline first so output/cleaned__targe

spark = (
    SparkSession.builder.appName("target-pipeline-eda")
    .master("local[*]")
    .config("spark.sql.warehouse.dir", str(OUTPUT_DIR / "spark-warehouse"))
    .enableHiveSupport()
    .getOrCreate()
)

spark_df = spark.read.option("header", True).option("inferSchema", True).csv

spark_df = (
    spark_df
    .withColumn("Id", F.col("Id").cast("int"))
    .withColumn("order_products_value", F.col("order_products_value").cast("
    .withColumn("order_freight_value", F.col("order_freight_value").cast("do
    .withColumn("order_items_qty", F.col("order_items_qty").cast("int"))
    .withColumn("customer_zip_code_prefix", F.col("customer_zip_code_prefix"
    .withColumn("review_score", F.col("review_score").cast("double"))
    .withColumn("total_order_value", F.col("total_order_value").cast("double
    .withColumn("approval_delay_hours", F.col("approval_delay_hours").cast("
    .withColumn("delivery_time_days", F.col("delivery_time_days").cast("doub
    .withColumn("freight_ratio", F.col("freight_ratio").cast("double"))
    .withColumn("order_purchase_timestamp", F.to_timestamp("order_purchase_t
    .withColumn("order_aproved_at", F.to_timestamp("order_aproved_at"))
```

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```
    .withColumn("order_delivered_customer_date", F.to_timestamp("order_deliv
    .withColumn("order_date", F.to_date("order_purchase_timestamp"))
    .withColumn("order_year", F.year("order_purchase_timestamp"))
    .withColumn("order_week_start", F.date_trunc("week", F.col("order_purcha
)

spark_df.createOrReplaceTempView("target_cleaned")
print(f"Loaded {spark_df.count()} cleaned rows")
```

```
26/03/27 00:38:11 WARN SparkSession: Using an existing Spark session; only r
untime SQL configurations will take effect.
Loaded 978 cleaned rows
```

# Dataset Shape and Schema Check

In [6]:
```
spark_df.printSchema()
```

```
root
 |-- Id: integer (nullable = true)
 |-- order_status: string (nullable = true)
 |-- order_products_value: double (nullable = true)
 |-- order_freight_value: double (nullable = true)
 |-- order_items_qty: integer (nullable = true)
 |-- order_purchase_timestamp: timestamp (nullable = true)
 |-- order_aproved_at: timestamp (nullable = true)
 |-- order_delivered_customer_date: timestamp (nullable = true)
 |-- customer_city: string (nullable = true)
 |-- customer_state: string (nullable = true)
 |-- customer_zip_code_prefix: integer (nullable = true)
 |-- review_score: double (nullable = true)
 |-- total_order_value: double (nullable = true)
 |-- approval_delay_hours: double (nullable = true)
 |-- delivery_time_days: double (nullable = true)
 |-- freight_ratio: double (nullable = true)
 |-- order_date: date (nullable = true)
 |-- order_year: integer (nullable = true)
 |-- order_week_start: timestamp (nullable = true)
```

In [7]:
```
schema_summary_df = spark.createDataFrame(
    [(field.name, field.dataType.simpleString(), spark_df.filter(F.col(field
    ["column", "dtype", "non_null_count"],
)
schema_summary_df.toPandas()
```

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Out[7]:

| | column | dtype | non_null_count |
|---|---|---|---|
| 0 | Id | int | 978 |
| 1 | order_status | string | 978 |
| 2 | order_products_value | double | 978 |
| 3 | order_freight_value | double | 978 |
| 4 | order_items_qty | int | 978 |
| 5 | order_purchase_timestamp | timestamp | 978 |
| 6 | order_aproved_at | timestamp | 978 |
| 7 | order_delivered_customer_date | timestamp | 978 |
| 8 | customer_city | string | 978 |
| 9 | customer_state | string | 978 |
| 10 | customer_zip_code_prefix | int | 978 |
| 11 | review_score | double | 978 |
| 12 | total_order_value | double | 978 |
| 13 | approval_delay_hours | double | 978 |
| 14 | delivery_time_days | double | 978 |
| 15 | freight_ratio | double | 978 |
| 16 | order_date | date | 978 |
| 17 | order_year | int | 978 |
| 18 | order_week_start | timestamp | 978 |

# EDA Checklist

In [9]:

```
eda_summary_df = spark.sql(
    """
    SELECT 'row_count' AS metric, CAST(COUNT(*) AS DOUBLE) AS value FROM tar
    UNION ALL
    SELECT 'mean_order_products_value', AVG(order_products_value) FROM targe
    UNION ALL
    SELECT 'mean_order_freight_value', AVG(order_freight_value) FROM target_
    UNION ALL
    SELECT 'unique_states', CAST(COUNT(DISTINCT customer_state) AS DOUBLE) F
    UNION ALL
    SELECT 'average_delivery_time_days', AVG(delivery_time_days) FROM target
    UNION ALL
    SELECT 'average_approval_delay_hours', AVG(approval_delay_hours) FROM ta
    UNION ALL
    SELECT 'average_review_score', AVG(review_score) FROM target_cleaned
```

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```
)
eda_summary_df.toPandas()
```

Out[9]:

| | metric | value |
|---|---|---|
| 0 | row_count | 978.00 |
| 1 | mean_order_products_value | 127.85 |
| 2 | mean_order_freight_value | 21.41 |
| 3 | unique_states | 26.00 |
| 4 | average_delivery_time_days | 12.33 |
| 5 | average_approval_delay_hours | 10.41 |
| 6 | average_review_score | 4.09 |

In [10]:

```
missing_values_exprs = [
    F.sum(F.when(F.col(column).isNull(), 1).otherwise(0)).alias(column)
    for column in spark_df.columns
]
missing_values_row = spark_df.select(missing_values_exprs).collect()[0].asDi
missing_values_pd = pd.DataFrame(
    [(column, missing_values_row[column]) for column in spark_df.columns],
    columns=["column", "missing_values"],
).sort_values(["missing_values", "column"], ascending=[False, True])
missing_values_pd
```

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Out[10]:

| | column | missing_values |
|---|---|---|
| 0 | Id | 0 |
| 13 | approval_delay_hours | 0 |
| 8 | customer_city | 0 |
| 9 | customer_state | 0 |
| 10 | customer_zip_code_prefix | 0 |
| 14 | delivery_time_days | 0 |
| 15 | freight_ratio | 0 |
| 6 | order_aproved_at | 0 |
| 16 | order_date | 0 |
| 7 | order_delivered_customer_date | 0 |
| 3 | order_freight_value | 0 |
| 4 | order_items_qty | 0 |
| 2 | order_products_value | 0 |
| 5 | order_purchase_timestamp | 0 |
| 1 | order_status | 0 |
| 18 | order_week_start | 0 |
| 17 | order_year | 0 |
| 11 | review_score | 0 |
| 12 | total_order_value | 0 |

In [11]:

```
states_df = spark_df.select("customer_state").distinct().orderBy("customer_s
print(f"Unique states: {states_df.count()}")
states_df.toPandas().T
```

```
Unique states: 26
```

Out[11]:

| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | ... | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **customer_state** | AC | AL | AM | BA | CE | DF | ES | GO | MA | MG | ... | PR | RJ | RN | RO | RR |

1 rows × 26 columns

In [12]:

```
total_orders = spark_df.count()
status_distribution_df = (
    spark_df.groupBy("order_status")
    .agg(F.count("Id").alias("order_count"))
    .withColumn(
        "order_percent",
        F.round(F.col("order_count") / F.lit(total_orders) * 100, 2),
```

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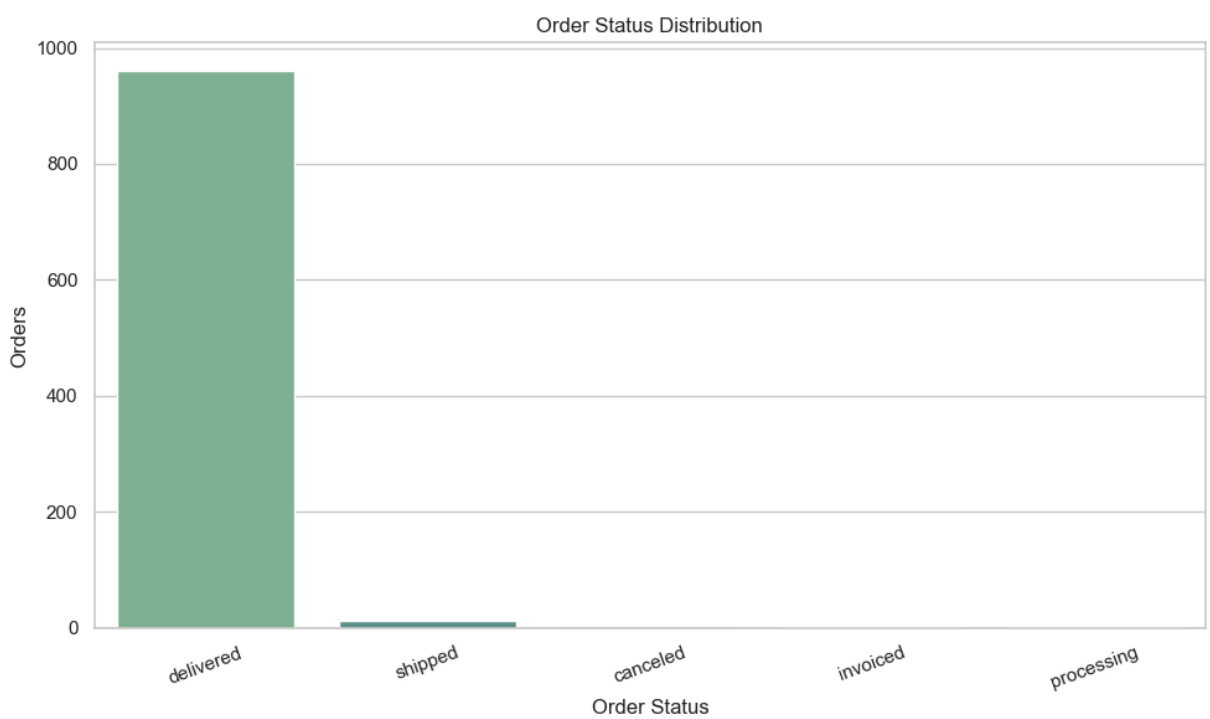
```
    .orderBy(F.desc("order_count"))
)
status_distribution_pd = status_distribution_df.toPandas()
status_distribution_pd
```

Out[12]:

| | order_status | order_count | order_percent |
|---|---|---|---|
| **0** | delivered | 961 | 98.26 |
| **1** | shipped | 12 | 1.23 |
| **2** | canceled | 3 | 0.31 |
| **3** | invoiced | 1 | 0.10 |
| **4** | processing | 1 | 0.10 |

In [13]:

```
plt.figure(figsize=(10, 6))
sns.barplot(
    data=status_distribution_pd,
    x="order_status",
    y="order_count",
    hue="order_status",
    dodge=False,
    palette="crest",
    legend=False,
)
plt.title("Order Status Distribution")
plt.xlabel("Order Status")
plt.ylabel("Orders")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```

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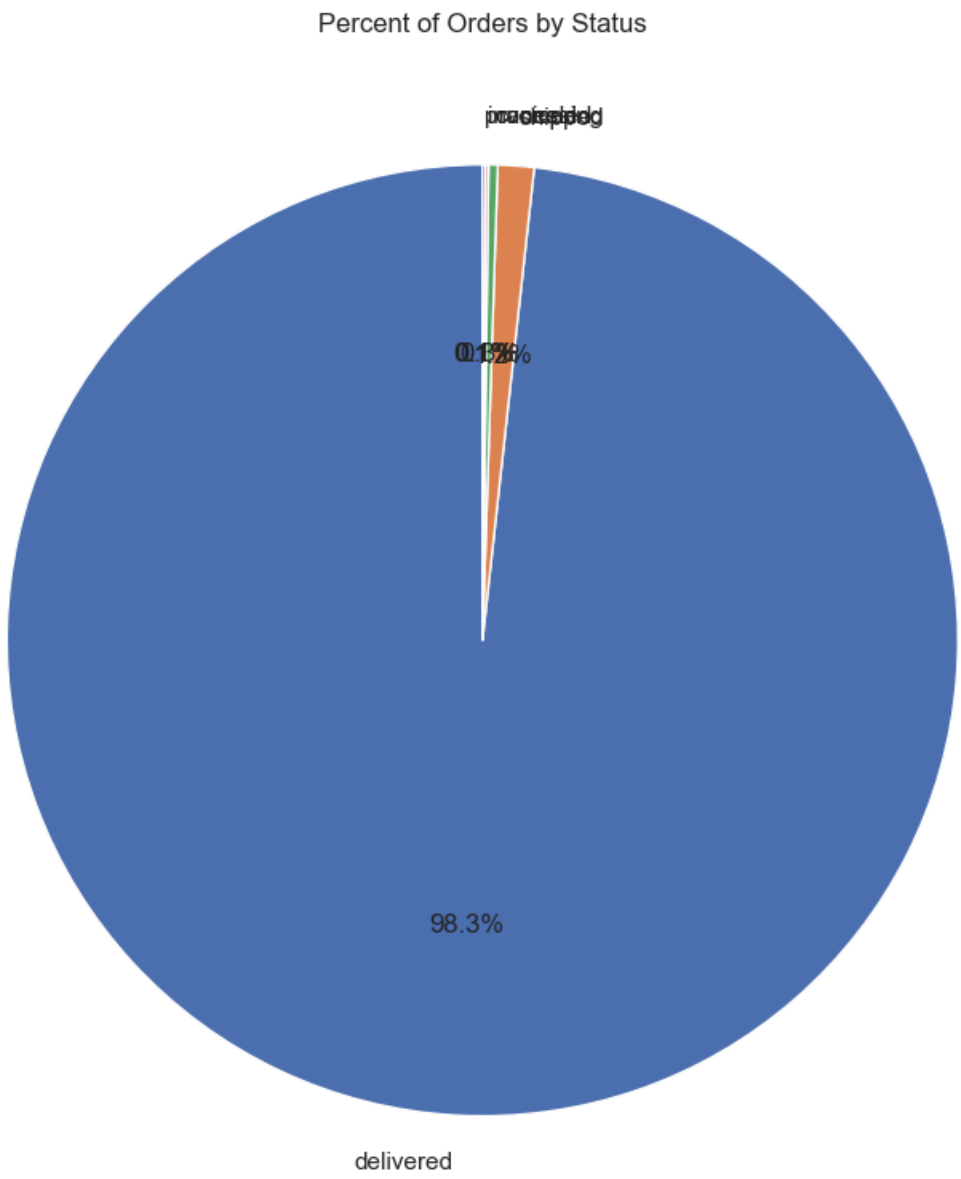
In [14]:
```
plt.figure(figsize=(8, 8))
plt.pie(
    status_distribution_pd["order_count"],
    labels=status_distribution_pd["order_status"],
    autopct="%1.1f%%",
    startangle=90,
)
plt.title("Percent of Orders by Status")
plt.tight_layout()
plt.show()
```




In [15]:
```
top_order_cities_df = (
    spark_df.groupBy("customer_city", "customer_state")
    .agg(F.count("Id").alias("order_count"))
    .orderBy(F.desc("order_count"), F.asc("customer_city"))
    .limit(5)
```

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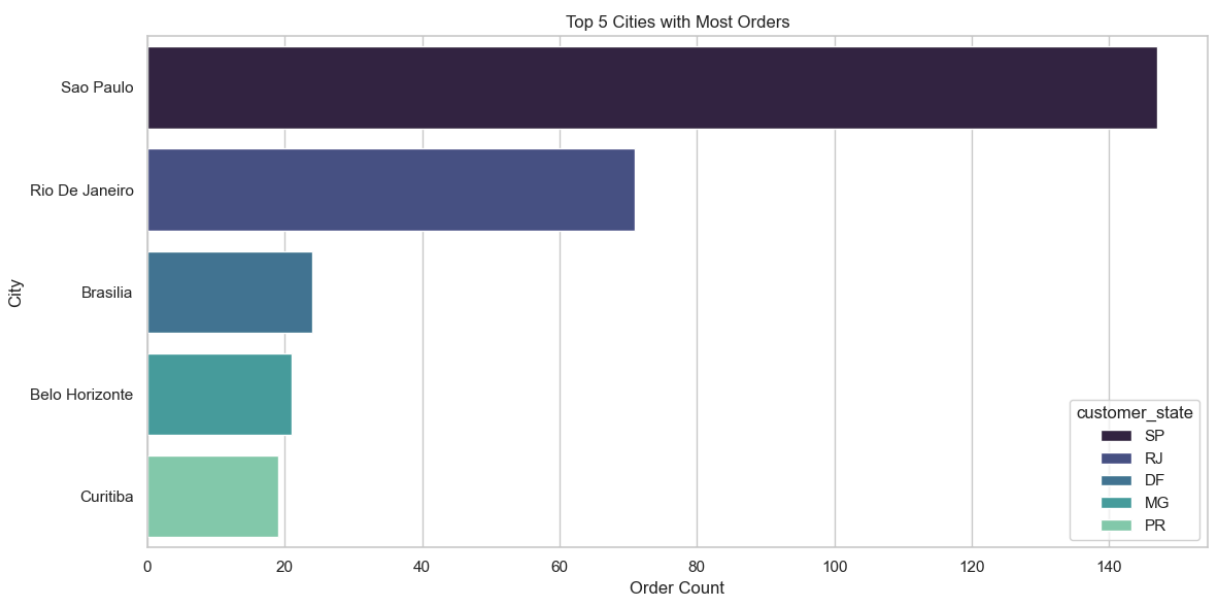
```
top_order_cities_pd = top_order_cities_df.toPandas()
top_order_cities_pd
```

Out[15]:

| | customer_city | customer_state | order_count |
|---|---|---|---|
| **0** | Sao Paulo | SP | 147 |
| **1** | Rio De Janeiro | RJ | 71 |
| **2** | Brasilia | DF | 24 |
| **3** | Belo Horizonte | MG | 21 |
| **4** | Curitiba | PR | 19 |

In [16]:

```
plt.figure(figsize=(12, 6))
sns.barplot(
    data=top_order_cities_pd,
    x="order_count",
    y="customer_city",
    hue="customer_state",
    dodge=False,
    palette="mako",
)
plt.title("Top 5 Cities with Most Orders")
plt.xlabel("Order Count")
plt.ylabel("City")
plt.tight_layout()
plt.show()
```




# Analysis Checklist

In [18]:

```
sales_by_city_df = (
    spark_df.groupBy("customer_city", "customer_state")
    .agg(F.round(F.sum("total_order_value"), 2).alias("total_sales"))
    .orderBy(F.desc("total_sales"))
```

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```
)
sales_by_city_pd = sales_by_city_df.limit(15).toPandas()
sales_by_city_df.limit(10).toPandas()
```

Out[18]:

| | customer_city | customer_state | total_sales |
|---|---|---|---|
| 0 | Sao Paulo | SP | 18,825.81 |
| 1 | Rio De Janeiro | RJ | 12,868.05 |
| 2 | Belo Horizonte | MG | 4,084.15 |
| 3 | Brasilia | DF | 2,652.86 |
| 4 | Curitiba | PR | 2,607.10 |
| 5 | Diadema | SP | 1,855.75 |
| 6 | Sao Goncalo | RJ | 1,621.56 |
| 7 | Fortaleza | CE | 1,601.98 |
| 8 | Limeira | SP | 1,573.05 |
| 9 | Santo Andre | SP | 1,531.92 |

In [19]:

```
plt.figure(figsize=(12, 8))
sns.barplot(
    data=sales_by_city_pd,
    x="total_sales",
    y="customer_city",
    hue="customer_state",
    dodge=False,
    palette="flare",
)
plt.title("Top 15 Cities by Total Sales")
plt.xlabel("Total Sales")
plt.ylabel("City")
plt.tight_layout()
plt.show()
```

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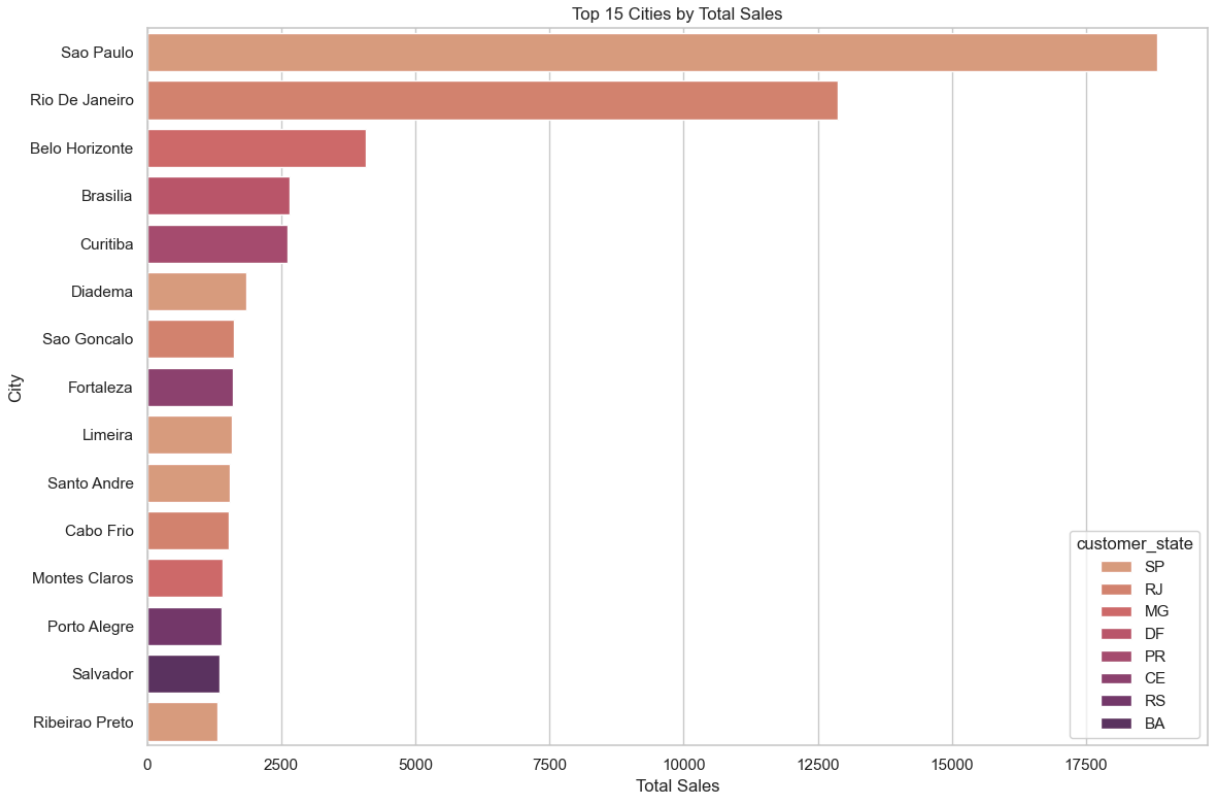


In [20]:
```
correlation_pairs = [
    ("order_products_value", "order_freight_value"),
    ("order_products_value", "order_items_qty"),
    ("order_freight_value", "order_items_qty"),
    ("delivery_time_days", "review_score"),
]
correlation_results = []
for left, right in correlation_pairs:
    correlation_results.append((left, right, spark_df.stat.corr(left, right)

correlation_pd = pd.DataFrame(correlation_results, columns=["left_column", "
correlation_pd["correlation"] = correlation_pd["correlation"].round(4)
correlation_pd
```

Out[20]:

| | left_column | right_column | correlation |
|---|---|---|---|
| **0** | order_products_value | order_freight_value | 0.47 |
| **1** | order_products_value | order_items_qty | 0.26 |
| **2** | order_freight_value | order_items_qty | 0.63 |
| **3** | delivery_time_days | review_score | 0.02 |

In [21]:
```
correlation_matrix_pd = pd.DataFrame(
    {
        "order_products_value": [1.0, spark_df.stat.corr("order_products_val
        "order_freight_value": [spark_df.stat.corr("order_products_value", "
        "order_items_qty": [spark_df.stat.corr("order_products_value", "orde
    }
    index=["order_products_value", "order_freight_value", "order_items_qty"]
```

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```
).round(4)

plt.figure(figsize=(7, 5))
sns.heatmap(correlation_matrix_pd, annot=True, cmap="coolwarm", fmt=".2f", v
plt.title("Correlation: Product Value, Freight, Item Quantity")
plt.tight_layout()
plt.show()
```

In [22]:
```
timing_summary_df = spark_df.agg(
    F.round(F.avg("delivery_time_days"), 2).alias("average_delivery_time_day
    F.round(F.avg("approval_delay_hours"), 2).alias("average_approval_delay_
)
timing_summary_df.toPandas()
```

Out[22]:

| | average_delivery_time_days | average_approval_delay_hours |
|---|---|---|
| **0** | 12.33 | 10.41 |

In [23]:
```
average_review_per_order_df = (
    spark_df.groupBy("Id")
    .agg(F.round(F.avg("review_score"), 2).alias("average_review_score"))
    .orderBy(F.desc("average_review_score"), F.asc("Id"))
)
average_review_per_order_df.limit(10).toPandas()
```

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Out[23]:

| | Id | average_review_score |
|---|---|---|
| **0** | 1 | 5.00 |
| **1** | 2 | 5.00 |
| **2** | 5 | 5.00 |
| **3** | 6 | 5.00 |
| **4** | 7 | 5.00 |
| **5** | 9 | 5.00 |
| **6** | 10 | 5.00 |
| **7** | 15 | 5.00 |
| **8** | 18 | 5.00 |
| **9** | 21 | 5.00 |

In [24]:
```
average_review_score_value = average_review_per_order_df.agg(F.round(F.avg("
print(f"Average review score per order: {average_review_score_value}")
```

```
Average review score per order: 4.09
```

In [25]:
```
delivery_by_city_df = (
    spark_df.groupBy("customer_city", "customer_state")
    .agg(
        F.round(F.avg("delivery_time_days"), 2).alias("average_delivery_time
        F.count("Id").alias("order_count"),
    )
    .filter(F.col("order_count") >= 3)
)
fastest_cities_pd = delivery_by_city_df.orderBy(F.asc("average_delivery_time
slowest_cities_pd = delivery_by_city_df.orderBy(F.desc("average_delivery_tim
print("Top 3 fastest delivery cities")
display(fastest_cities_pd)
print("Top 3 slowest delivery cities")
display(slowest_cities_pd)
```

```
Top 3 fastest delivery cities
```

| | customer_city | customer_state | average_delivery_time_days | order_count |
|---|---|---|---|---|
| **0** | Cabo Frio | RJ | 5.40 | 3 |
| **1** | Pelotas | RS | 6.30 | 3 |
| **2** | Ipatinga | MG | 6.92 | 3 |

```
Top 3 slowest delivery cities
```

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| | customer_city | customer_state | average_delivery_time_days | order_count |
|---|---|---|---|---|
| **0** | Contagem | MG | 24.64 | 3 |
| **1** | Orlandia | SP | 22.33 | 3 |
| **2** | Teresina | PI | 20.08 | 3 |

In [26]:
```
review_delivery_df = (
    spark_df.groupBy("review_score")
    .agg(
        F.round(F.avg("delivery_time_days"), 2).alias("average_delivery_time
        F.count("Id").alias("order_count"),
    )
    .orderBy("review_score")
)
review_delivery_pd = review_delivery_df.toPandas()
review_delivery_pd
```

Out[26]:

| | review_score | average_delivery_time_days | order_count |
|---|---|---|---|
| **0** | 1.00 | 11.92 | 115 |
| **1** | 2.00 | 11.03 | 25 |
| **2** | 3.00 | 12.82 | 80 |
| **3** | 4.00 | 12.06 | 193 |
| **4** | 5.00 | 12.49 | 565 |

In [27]:
```
plt.figure(figsize=(10, 6))
sns.lineplot(data=review_delivery_pd, x="review_score", y="average_delivery_
plt.title("Average Delivery Time by Review Score")
plt.xlabel("Review Score")
plt.ylabel("Average Delivery Time (Days)")
plt.tight_layout()
plt.show()
```

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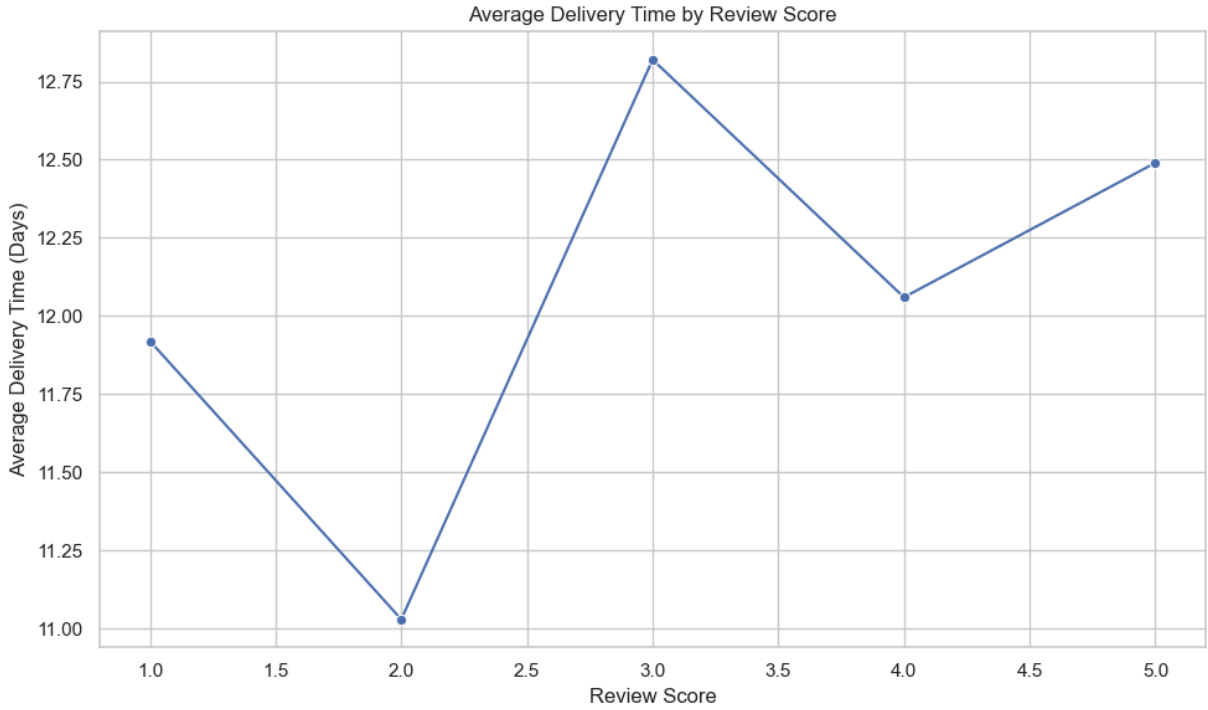


In [28]:
```
review_scatter_pd = spark_df.select("delivery_time_days", "review_score").to
plt.figure(figsize=(10, 6))
sns.regplot(
    data=review_scatter_pd,
    x="delivery_time_days",
    y="review_score",
    scatter_kws={"alpha": 0.35, "s": 35},
    line_kws={"color": "#c44e52"},
)
plt.title("Review Score vs Delivery Time")
plt.xlabel("Delivery Time (Days)")
plt.ylabel("Review Score")
plt.tight_layout()
plt.show()
```

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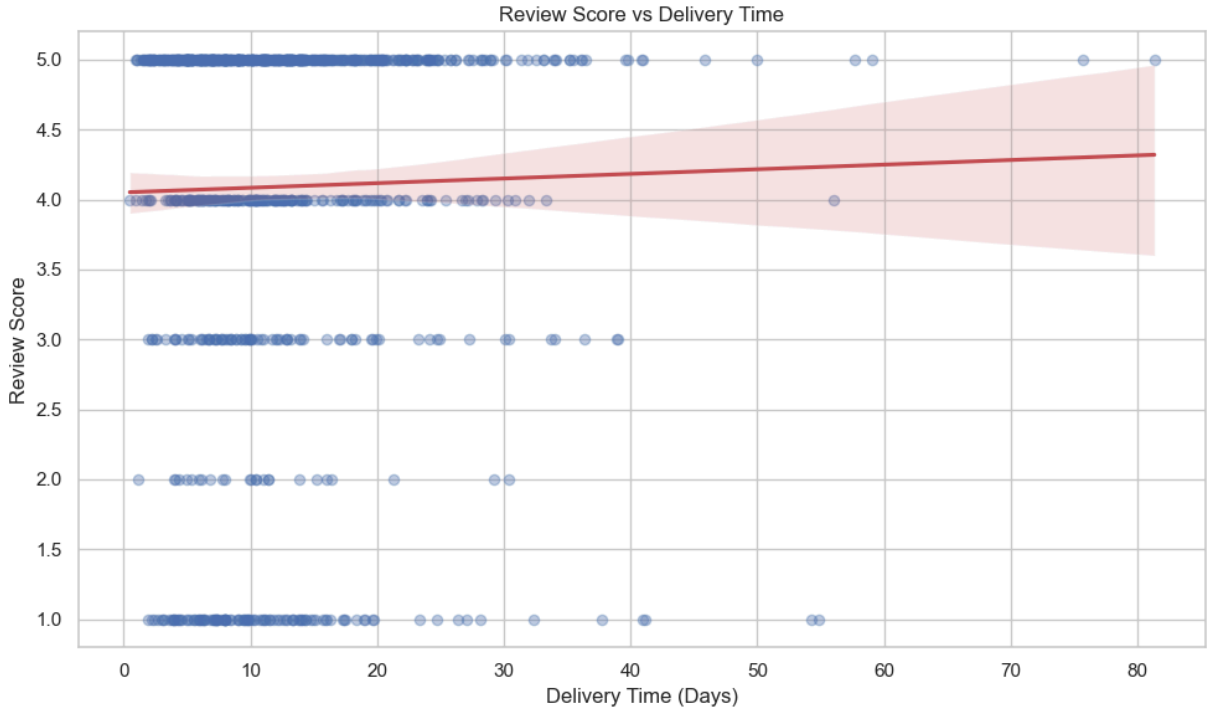


# Daily and Weekly Sales Views

In [30]:
```
daily_sales_df = (
    spark_df.groupBy("order_date")
    .agg(F.round(F.sum("total_order_value"), 2).alias("total_sales"))
    .orderBy("order_date")
)
daily_sales_pd = daily_sales_df.toPandas()
daily_sales_pd.head(10)
```

Out[30]:

| | order_date | total_sales |
|---|---|---|
| 0 | 2016-10-05 | 318.33 |
| 1 | 2017-01-05 | 162.78 |
| 2 | 2017-01-14 | 44.57 |
| 3 | 2017-01-21 | 83.28 |
| 4 | 2017-01-22 | 159.37 |
| 5 | 2017-01-23 | 111.13 |
| 6 | 2017-01-25 | 31.78 |
| 7 | 2017-01-26 | 101.93 |
| 8 | 2017-01-27 | 262.94 |
| 9 | 2017-01-28 | 53.37 |

In [31]:
```
plt.figure(figsize=(14, 6))
sns.lineplot(data=daily_sales_pd, x="order_date", y="total_sales")
```

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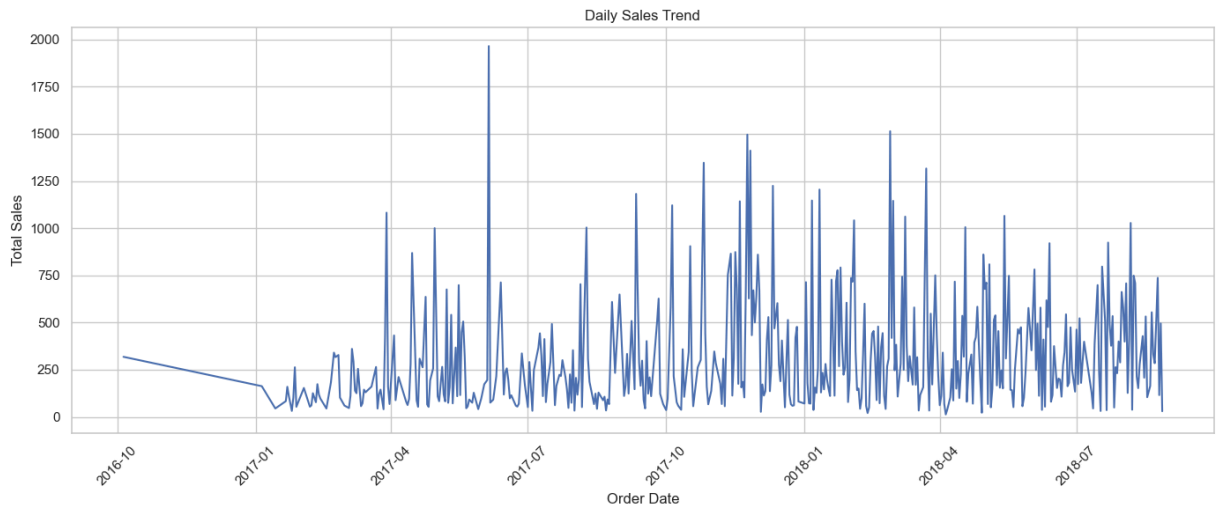
```
plt.title("Daily Sales Trend")
plt.xlabel("Order Date")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```




In [32]:
```
weekly_sales_df = (
    spark_df.groupBy("order_week_start")
    .agg(F.round(F.sum("total_order_value"), 2).alias("total_sales"))
    .orderBy("order_week_start")
)
weekly_sales_pd = weekly_sales_df.toPandas()
weekly_sales_pd.head(10)
```

Out[32]:

| | order_week_start | total_sales |
|---|---|---|
| 0 | 2016-10-03 | 318.33 |
| 1 | 2017-01-02 | 162.78 |
| 2 | 2017-01-09 | 44.57 |
| 3 | 2017-01-16 | 242.65 |
| 4 | 2017-01-23 | 561.15 |
| 5 | 2017-01-30 | 153.26 |
| 6 | 2017-02-06 | 609.87 |
| 7 | 2017-02-13 | 141.18 |
| 8 | 2017-02-20 | 1,270.35 |
| 9 | 2017-02-27 | 217.66 |

In [33]:
```
plt.figure(figsize=(14, 6))
sns.lineplot(data=weekly_sales_pd, x="order_week_start", y="total_sales", ma
plt.title("Weekly Sales Trend")
plt.xlabel("Week Start")
plt.ylabel("Total Sales")
```

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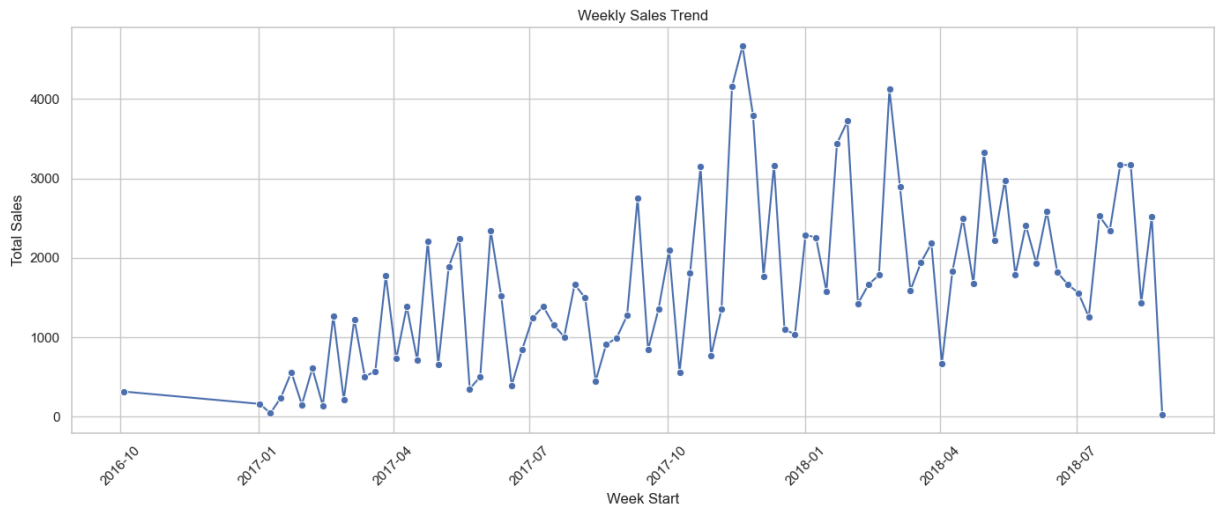
```
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```




# Product Orders by Year

In [35]:
```
products_per_year_df = (
    spark_df.groupBy("order_year")
    .agg(F.sum("order_items_qty").alias("order_items_qty"))
    .orderBy("order_year")
)
products_per_year_pd = products_per_year_df.toPandas()
products_per_year_pd
```

Out[35]:

| | order_year | order_items_qty |
|---|---|---|
| **0** | 2016 | 2 |
| **1** | 2017 | 498 |
| **2** | 2018 | 567 |

In [36]:
```
plt.figure(figsize=(10, 6))
sns.barplot(
    data=products_per_year_pd,
    x="order_year",
    y="order_items_qty",
    hue="order_year",
    dodge=False,
    palette="rocket",
    legend=False,
)
plt.title("Total Products Ordered per Year")
plt.xlabel("Year")
plt.ylabel("Sum of Ordered Products")
plt.tight_layout()
plt.show()
```

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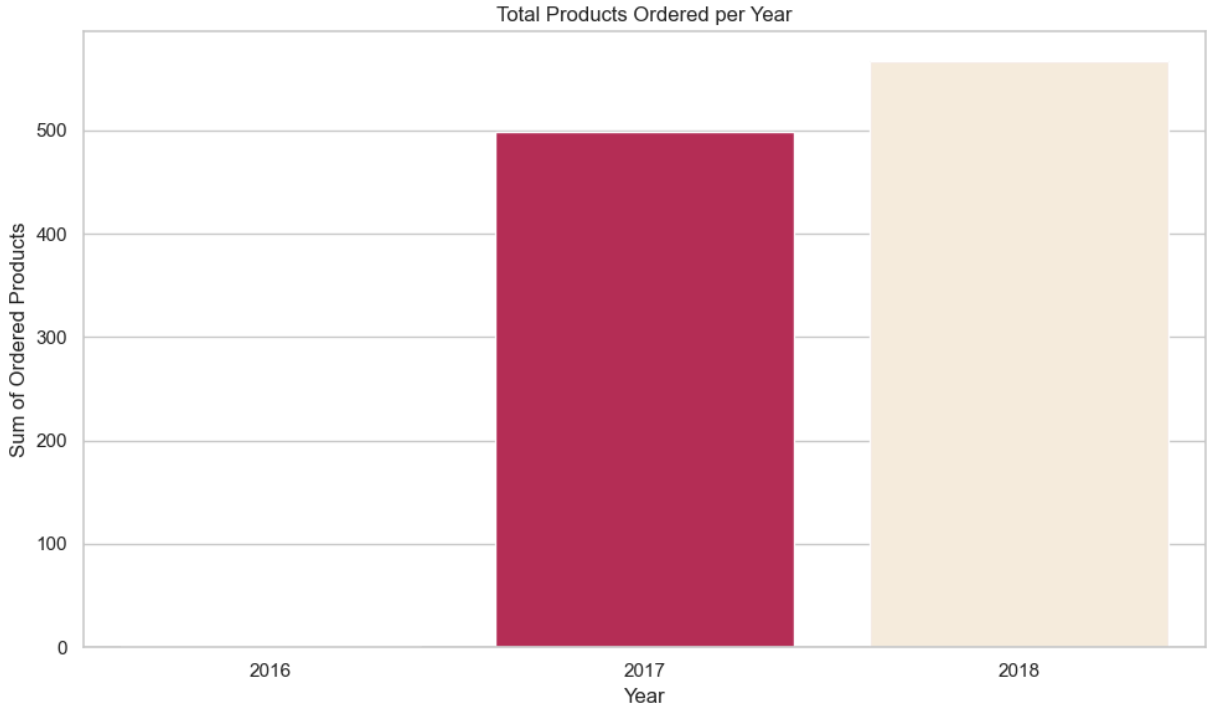


## Spark Output Check

In [38]:
```
city_report_paths = sorted(CITY_REPORT_DIR.glob("part-*.csv"))
if city_report_paths:
    city_report_spark_df = spark.read.option("header", True).option("inferSc
    city_report_pd = city_report_spark_df.orderBy(F.desc("total_sales")).lim
    display(city_report_spark_df.limit(10).toPandas())
    plt.figure(figsize=(12, 8))
    sns.barplot(
        data=city_report_pd,
        x="total_sales",
        y="customer_city",
        hue="customer_state",
        dodge=False,
        palette="viridis",
    )
    plt.title("Top 15 City-State Sales from Spark Report")
    plt.xlabel("Total Sales")
    plt.ylabel("City")
    plt.legend(title="State", bbox_to_anchor=(1.02, 1), loc="upper left")
    plt.tight_layout()
    plt.show()
else:
    print("No Spark city report found under output/sales_by_city_report/")
```

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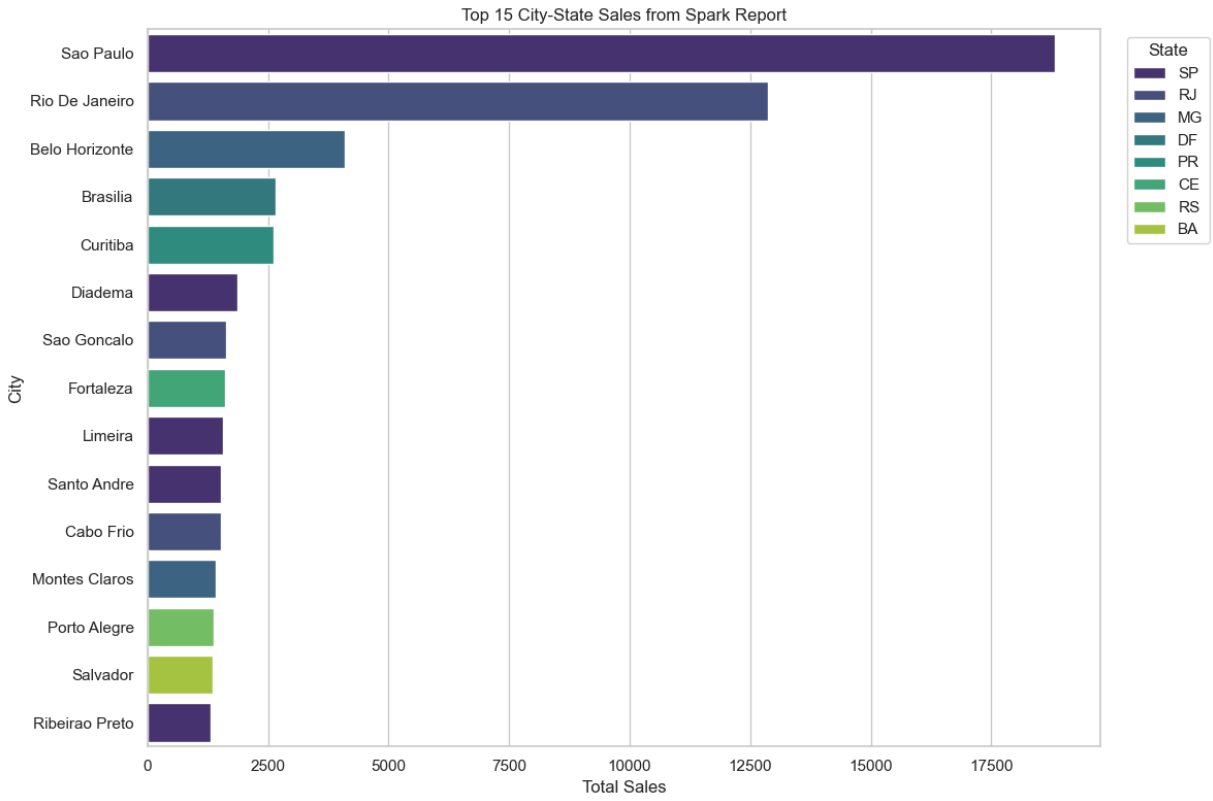
| | customer_city | customer_state | total_sales | avg_delivery_days | avg_review_score |
|---|---|---|---|---|---|
| **0** | Sao Paulo | SP | 18,825.81 | 13.00 | 4.24 |
| **1** | Rio De Janeiro | RJ | 12,868.05 | 12.59 | 4.13 |
| **2** | Belo Horizonte | MG | 4,084.15 | 12.57 | 3.95 |
| **3** | Brasilia | DF | 2,652.86 | 10.72 | 3.79 |
| **4** | Curitiba | PR | 2,607.10 | 13.74 | 3.95 |
| **5** | Diadema | SP | 1,855.75 | 7.33 | 3.00 |
| **6** | Sao Goncalo | RJ | 1,621.56 | 13.41 | 3.86 |
| **7** | Fortaleza | CE | 1,601.98 | 10.73 | 4.22 |
| **8** | Limeira | SP | 1,573.05 | 14.19 | 3.67 |
| **9** | Santo Andre | SP | 1,531.92 | 11.99 | 4.64 |

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